



GOSFORD HIGH SCHOOL

2013

HSC ASSESSMENT TASK 3

EXTENSION 2 MATHEMATICS

General Instructions:

Attempt all Questions

- Working time: 60 minutes
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question in Section 2 should be Started on a separate page.
- All necessary working should be shown in every question.

Total marks: - 40

SECTION 1	/ 4
SECTION 2 Q1	/ 12
SECTION 2 Q2	/ 12
SECTION 2 Q3	/ 12
TOTAL	/ 40

SECTION 1. (4 MARKS) Answer on the answer sheet provided.

Question 1.

If $z = 1 + i$ is a root of $z^2 - aiz + b = 0$, where a and b are real numbers then

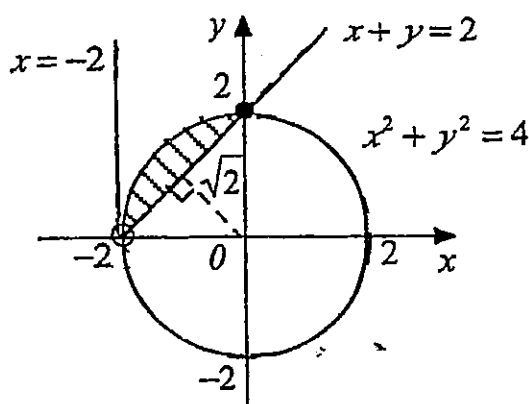
- | | |
|---------------------|--------------------|
| A. $a = 2, b = -2$ | C. $a = 2, b = 2$ |
| B. $a = -2, b = -2$ | D. $a = -2, b = 2$ |

Question 2.

If $z = \sqrt{2} - i\sqrt{2}$, then z^5 written in mod-arg form is

- | | |
|--|---|
| A. $2\sqrt{2} \operatorname{cis} \frac{-\pi}{4}$ | C. $32 \operatorname{cis} \frac{-\pi}{4}$ |
| B. $2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}$ | D. $32 \operatorname{cis} \frac{3\pi}{4}$ |

Question 3.



The shaded region in the Argand diagram above shows all the points representing complex numbers z such that

- | |
|---|
| A. $ z \leq 2$ and $\frac{\pi}{4} \leq \arg(z - 2) \leq \frac{\pi}{2}$. |
| B. $ z \leq 2$ and $\frac{\pi}{4} \leq \arg(z + 2) \leq \frac{\pi}{2}$. |
| C. $ z \geq 2$ and $\frac{\pi}{4} \leq \arg(z - 2) \leq \frac{\pi}{2}$. |
| D. $ z \geq 2$ and $\frac{\pi}{4} \leq \arg(z + 2) \leq \frac{\pi}{2}$. |

Question 4.

If $\left| \frac{z-1}{z} \right| = 2$ then the locus of z is given by

- | | |
|-------------------------------|-----------------------------|
| A. $3x^2 + 3y^2 + 2x - 1 = 0$ | C. $x^2 + y^2 + 2x - 1 = 0$ |
| B. $3x^2 + 3y^2 - 2x - 1 = 0$ | D. $x^2 + y^2 - 2x + 1 = 0$ |



Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐

B ☒

C ☐

D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒

B ☒

C ☐

D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒

B ☒

C ☐

D ☐

correct



Start
here



1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

HSC ASSESSMENT TASK 3

EXTENSION 2 MATHEMATICS

SECTION 2. (36 marks) Answer on your own paper. Show all necessary working

Question 1. (12 marks) **START A NEW PAGE.**

(a) Consider the equation $z^3 + mz^2 + nz + 6 = 0$, where m and n are real. It is known that $1 - i$ is a root of this equation.

(i) Find the other roots of the equation. (2)

(ii) Find the values of m and n . (2)

(b) α , β , and γ are roots of the equation $x^3 - 6x^2 + 12x - 35 = 0$.

(i) Form a cubic equation whose roots are $\alpha - 2, \beta - 2, \gamma - 2$. (2)

(ii) Hence, or otherwise solve the equation $x^3 - 6x^2 + 12x - 35 = 0$ over the complex field. (2)

(c) Solve the equation $x^4 - 6x^2 + 8x - 3 = 0$, given that it has a triple root (4)

Question 2. (12 marks) **START A NEW PAGE.**

(a) Find (i) $\int \frac{\ln x}{x} dx$ (1)

(ii) $\int \frac{\ln x}{x^2} dx$ (3)

(b) (i) If $\frac{x-11}{x^2+3x-4} = \frac{A}{x+4} + \frac{B}{x-1}$, find A and B . (2)

(ii) Hence, evaluate $\int_2^3 \frac{x-11}{x^2+3x-4} dx$. (2)

(c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2+\cos \theta}$. (4)

Question 3. (12 marks) **START A NEW PAGE.**

(a) Find $\int \frac{x^2}{1+9x^2} dx$. (3)

(b) Use the substitution $x = \tan \theta$ to evaluate $\int_1^{\sqrt{3}} \frac{dx}{x^2\sqrt{1+x^2}}$ (4)

(c) (i) If $I_n = \int_{-1}^0 x^n \sqrt{1+x} dx$, for $n = 0, 1, 2, \dots$ show that

$$I_n = \frac{-2n}{2n+3} I_{n-1}, \quad \text{for } n = 1, 2, 3, \dots \quad (3)$$

(ii) Hence evaluate $\int_{-1}^0 x^3 \sqrt{1+x} dx$. (2)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

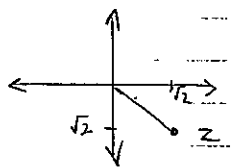
NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONS

Section 1

1. If $z = 1+i$ is a root
 $(1+i)^2 - a(1+i) + b = 0 + 0i$
 $1 + 2i + i^2 - a - ai + b = 0 + 0i$
 $1 + 2i - 1 - a - ai + b = 0 + 0i$
 $(a+b) + i(2-a) = 0 + 0i$
 $\therefore 2-a = 0$
 $a = 2$
 $b = -2$
 Hence: (A)

2. $z = \sqrt{2} - i\sqrt{2}$



$\therefore z = 2 \operatorname{cis}(-\frac{\pi}{4})$

$z^5 = 2^5 \operatorname{cis}(-\frac{5\pi}{4})$
 $= 32 \operatorname{cis} \frac{3\pi}{4}$

Hence: (D)

3. Shaded region is inside $x^2 + y^2 = 4$

$\therefore |z| \leq 2$

Shaded region is above and to the left of $\arg z = \frac{\pi}{4}$

$\therefore \frac{\pi}{4} \leq \arg(z+2) \leq \frac{\pi}{2}$

Hence: (B)

4. $|\frac{z-1}{z}| = 2$

Square both sides

$|z-1|^2 = 4|z|^2$

Let $z = x+iy$

$|x+iy-1|^2 = 4|x+iy|^2$

$(x-1)^2 + y^2 = 4(x^2 + y^2)$

$x^2 - 2x + 1 + y^2 = 4x^2 + 4y^2$

$0 = 3x^2 + 3y^2 + 2x - 1$

Hence: (A)

Section 2

- 1/ (i) Since the coefficients are real, if $1-i$ is a root then so is $1+i$
 Let the third root be α

Now $\sum \alpha\beta\gamma = \alpha(1-i)(1+i)$
 $= \alpha(1-i^2)$
 $= 2\alpha$

$\therefore 2\alpha = -\frac{d}{a}$

$2\alpha = -6$

$\alpha = -3$

$\therefore$ the roots are $-3, 1+i, 1-i$

(2)

(iii) $\sum \alpha = -3 + 1+i + 1-i$
 $= -1$

$\therefore -1 = -\frac{b}{a}$

$-1 = -m$

$m = 1$

(1)

$\sum \alpha\beta = (1+i)(1-i) + -3(1-i) + -3(1+i)$
 $= 2 - 3 + 3i - 3 - 3i$
 $= -4$

$\therefore -4 = \frac{c}{a}$

$-4 = n$

$n = -4$

(1)

b)

(i) Let $y = x - 2$ i.e. $x = y + 2$

So $(y+2)^3 - 6(y+2)^2 + 12(y+2) - 35 = 0$
 $y^3 + 3(y^2)(2) + 3(y)(2)^2 + (2)^3 - 6[y^2 + 4y + 4] + 12y + 24 - 35 = 0$

i.e. $y^3 + 6y^2 + 12y + 8 - 6y^2 - 24y - 24 + 12y + 24 - 35 = 0$

$\therefore y^3 - 27 = 0$ (2)

(ii) If $y^3 - 27 = 0$
 $(y-3)(y^2 + 3y + 9) = 0$

$y = 3$ or $y = \frac{-3 \pm \sqrt{9-36}}{2}$

$y = 3$ or $\frac{-3 \pm 3\sqrt{3}i}{2}$

Now the roots of $x^3 - 6x^2 + 12x - 35 = 0$
 are two more than the roots of $y^3 - 27 = 0$

$\therefore$ the required roots are $5, \frac{1}{2} + \frac{3\sqrt{3}i}{2}, \frac{1}{2} - \frac{3\sqrt{3}i}{2}$ (3)

c) Let $P(x) = x^4 - 6x^2 + 8x - 3$

$P'(x) = 4x^3 - 12x + 8$

$P''(x) = 12x^2 - 12$

If $P''(x) = 0$

$12x^2 - 12 = 0$

$x^2 - 1 = 0$

$x = \pm 1$

Now $P'(1) = 4 - 12 + 8$
 $= 0$

$\therefore P(1) = 1 - 6 + 8 - 3$
 $= 0$

$\therefore x = 1$ is a triple root

Let the other root be α

So $P(x) = (x-1)^3(x-\alpha)$

Now $\sum \alpha \beta \gamma \delta = \frac{e}{a}$

(4)

$\therefore (1)^3 \alpha = -3$
 $\alpha = -3$

$\therefore x = 1$ is a triple root & -3 is a single root

2/ a) i) $\int \frac{\ln x}{x} dx = \int \frac{1}{x} \cdot \ln x dx$
 $= \frac{(\ln x)^2}{2} + c$ (1)

ii) $\int \frac{\ln x}{x^2} dx = I$

Using integration by parts

$u = \ln x, v' = x^{-2}$
 $u' = \frac{1}{x}, v = -\frac{1}{x}$

$I = -\frac{\ln x}{x} - \int -\frac{1}{x^2} dx$

$= -\frac{\ln x}{x} + \int x^{-2} dx$

$= -\frac{\ln x}{x} - x^{-1} + c$

$= -\frac{\ln x - 1}{x} + c$ (3)

$$b) (i) \frac{x-11}{x^2+3x-4} = \frac{x-11}{(x+4)(x-1)}$$

$$\therefore \frac{x-11}{(x+4)(x-1)} = \frac{A}{x+4} + \frac{B}{x-1}$$

$$\text{Hence } x-11 = A(x-1) + B(x+4)$$

$$\text{If } x=1, -10 = 5B$$

$$B = -2$$

$$\text{If } x=-4, -15 = -5A$$

$$A = 3$$

$$\frac{x-11}{(x+4)(x-1)} = \frac{3}{x+4} - \frac{2}{x-1}$$

(2)

$$\begin{aligned} (ii) \int_2^3 \frac{x-11}{x^2+3x-4} dx &= \int_2^3 \left(\frac{3}{x+4} - \frac{2}{x-1} \right) dx \\ &= \left[3 \ln(x+4) - 2 \ln(x-1) \right]_2^3 \\ &= (3 \ln 7 - 2 \ln 2) - (3 \ln 6 - 2 \ln 1) \\ &= 3 \ln 7 - 2 \ln 2 - 3 \ln 6 \\ &= \ln \left(\frac{343}{864} \right) \end{aligned}$$

(2)

$$c) \int_0^{\pi/2} \frac{d\theta}{2+\cos\theta}$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\therefore d\theta = \frac{2}{1+t^2} dt \quad (\text{working not needed})$$

$$\text{If } \theta=0, t=0$$

$$\theta=\frac{\pi}{2}, t=1$$

$$\therefore I = \int_0^1 \frac{1}{2+\frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1}{2+2t^2+1-t^2} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{3+t^2} dt$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right]$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right)$$

$$= \frac{\pi}{3\sqrt{3}} \quad \text{or} \quad \frac{\pi\sqrt{3}}{9}$$

(4)

$$\begin{aligned} 3) a) \int \frac{x^2}{1+9x^2} dx &= \frac{1}{9} \int \frac{9x^2}{1+9x^2} dx \\ &= \frac{1}{9} \int \frac{1+9x^2}{1+9x^2} - \frac{1}{1+9x^2} dx \\ &= \frac{1}{9} \int \left(1 - \frac{1}{1+9x^2} \right) dx \\ &= \frac{1}{9} \left\{ x - \frac{1}{9} \times \frac{1}{\frac{1}{3}} \tan^{-1} \frac{x}{\frac{1}{3}} \right\} + C \\ &= \frac{1}{9} \left\{ x - \frac{1}{3} \tan^{-1} 3x \right\} + C \end{aligned}$$

(3)

$$b) \int_1^{\sqrt{3}} \frac{dx}{x^2 \sqrt{1+x^2}}$$

$$\text{Let } x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$\text{If } x=1, \theta = \frac{\pi}{4}$$

$$\text{If } x=\sqrt{3}, \theta = \frac{\pi}{3}$$

$$\begin{aligned}
 \therefore \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{1+x^2}} &= \int_{\pi/4}^{\pi/2} \frac{\sec^2 \theta \, d\theta}{\tan^2 \theta \sqrt{1+\tan^2 \theta}} \\
 &= \int_{\pi/4}^{\pi/2} \frac{\sec^2 \theta \, d\theta}{\tan^2 \theta \cdot \sec \theta} \\
 &= \int_{\pi/4}^{\pi/2} \frac{d\theta}{\frac{\sin^2 \theta}{\cos^2 \theta} \times \sec \theta} \\
 &= \int_{\pi/4}^{\pi/2} \frac{\cos \theta}{\sin^2 \theta} \, d\theta \\
 &= \int_{\pi/4}^{\pi/2} \cot \theta \cdot \operatorname{cosec} \theta \, d\theta \quad (4) \\
 &= \left[-\operatorname{cosec} \theta \right]_{\pi/4}^{\pi/2} \\
 &= -\frac{2}{\sqrt{3}} - (-\sqrt{2}) \\
 &= \sqrt{2} - \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \sqrt{2} - \frac{2\sqrt{3}}{3} \quad \text{or} \quad \frac{3\sqrt{2} - 2\sqrt{3}}{3}
 \end{aligned}$$

c) (i) $I_n = \int_{-1}^0 x^n \sqrt{1+x} \, dx$, $n = 0, 1, 2, \dots$

$$\begin{aligned}
 u &= x^n & v' &= (1+x)^{1/2} \\
 u' &= nx^{n-1} & v &= \frac{2}{3} (1+x)^{3/2}
 \end{aligned}$$

$$\begin{aligned}
 I_n &= \left[x^n \cdot \frac{2}{3} (1+x)^{3/2} \right]_{-1}^0 - \int_{-1}^0 \frac{2}{3} (1+x)^{3/2} \cdot nx^{n-1} \, dx \\
 &= (0-0) - \frac{2n}{3} \int_{-1}^0 x^{n-1} (1+x)(1+x)^{1/2} \, dx
 \end{aligned}$$

$$= 0 - \frac{2n}{3} \int_{-1}^0 x^{n-1} \sqrt{1+x} + x^2 \sqrt{1+x} \, dx$$

$$I_n = -\frac{2n}{3} (I_{n-1} + I_n)$$

$$I_n + \frac{2n}{3} I_n = -\frac{2n}{3} I_{n-1}$$

$$\frac{3I_n + 2nI_n}{3} = -\frac{2nI_{n-1}}{3}$$

$$\begin{aligned}
 \therefore I_n(3+2n) &= -2n I_{n-1} \\
 I_n &= \frac{-2n}{2n+3} I_{n-1}
 \end{aligned}$$

(3)

(ii) $I_3 = -\frac{6}{9} I_2$

$$= -\frac{6}{9} \times -\frac{4}{7} \times I_1$$

$$= -\frac{6}{9} \times -\frac{4}{7} \times -\frac{2}{5} \times I_0$$

$$\begin{aligned}
 \text{Now } I_0 &= \int_{-1}^0 (1+x)^{1/2} \, dx \\
 &= \left[\frac{2}{3} (1+x)^{3/2} \times 1 \right]_{-1}^0 \\
 &= \frac{2}{3} - 0 \\
 &= \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 \therefore I_3 &= -\frac{6}{9} \times -\frac{4}{7} \times -\frac{2}{5} \times \frac{2}{3} \\
 &= -\frac{32}{315}
 \end{aligned}$$

(2)